



First Name: _____

Family Name: _____

Student Number: _____

Diploma of Engineering
University Entrance Program
Foundation Studies Program

DPEN022 – Enabling Mathematics B

Differentiations Exam: Sample

This exam represents 35% of the total marks for the subject

Time allowed:	1 hour and 20 minutes
Items permitted in exam:	UOW approved calculator
Aids supplied for exam:	Formula Sheet

INSTRUCTIONS TO STUDENTS

1. Write in blue or black pens only. Pencils are permitted for diagrams and graphs.
2. All questions are to be attempted.
3. All notation is as given in lectures.
4. Show all necessary WORKING and reasoning in the spaces provided.
5. You must achieve a minimum of 40% in the Differentiation Exam. If this requirement is not met, your final mark for the subject will be a Technical Fail.
6. Total number of questions: 18
 - Questions 1-6 are multiple choice.
 - Each question is worth 1 mark.
 - Marks are awarded for the answer only.
 - Circle the letter (A, B, C, D, or E) corresponding to the correct answer of each question.
 - Questions 7-18 are long answer questions.
 - The value of each question is indicated.
 - Part marks may be awarded.

No examination material may be removed from the examination venue.

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Questions 1-6 are multiple choice. Show all necessary WORKING and reasoning in the spaces provided. Circle the letter (A, B, C, D, or E) corresponding to the correct answer of each question.

Question 1 *[Limits]*

Evaluate the limit, if it exists: $\lim_{x \rightarrow -2} \frac{(x+2)^3}{x^2 + 2x + 5}$

- (A) 0
- (B) $-\infty$
- (C) -2
- (D) 4
- (E) Does not exist

Question 2 *[Derivative of powers of x]*

Given $f(x) = x^2 - 6x - 1$, find the value of x so that $f'(x) = 0$.

- (A) $x = -1$
- (B) $x = 6$
- (C) $x = -3$
- (D) $x = 3$
- (E) $x = 0$

Question 3 [*Derivative of powers of x using Rules 1-5*]

Find the gradient function of the curve $y = \frac{1}{\sqrt[3]{x^2 - 1}}$.

- (A) $-\frac{1}{3\sqrt[3]{(x^2 - 1)^4}}$
- (B) $\frac{2x}{3\sqrt[3]{(x^2 - 1)^4}}$
- (C) $-\frac{2x}{3\sqrt[3]{(x^2 - 1)^4}}$
- (D) $-\frac{2x}{\sqrt[3]{(x^2 - 1)}}$
- (E) $-\frac{2}{3\sqrt[3]{(x^2 - 1)^4}}$

Question 4 [*Derivative of any functions using rules 1-5*]

Find the derivative of $f(x) = \frac{3x^2 + 1}{x^3}$

- (A) $\frac{6x}{3x^2}$
- (B) $\frac{3}{x} - \frac{1}{x^3}$
- (C) $-\frac{6}{x^3} - \frac{3}{x^5}$
- (D) $\frac{3}{x^2} - \frac{3}{x^4}$
- (E) $\frac{-3}{x^2} - \frac{3}{x^4}$

Question 5 *[Mixed derivative of any functions using rules 1-5]*

Find the derivative of $y = \frac{\sin(3x)}{e^x}$.

- (A) $\frac{3e^x \cos(3x) - e^x \sin(3x)}{e^{2x}}$
- (B) $\frac{3 \cos(3x) - \sin(3x)}{e^{2x}}$
- (C) $\frac{3e^x \sin(3x) - \sin(3x)}{e^{2x}}$
- (D) $\frac{3e^x \cos(3x) + \sin(3x)}{e^{2x}}$
- (E) $\frac{3 \cos(3x) - e^x \sin(3x)}{e^{2x}}$

Question 6 *[Equation of tangent to function with powers of x]*

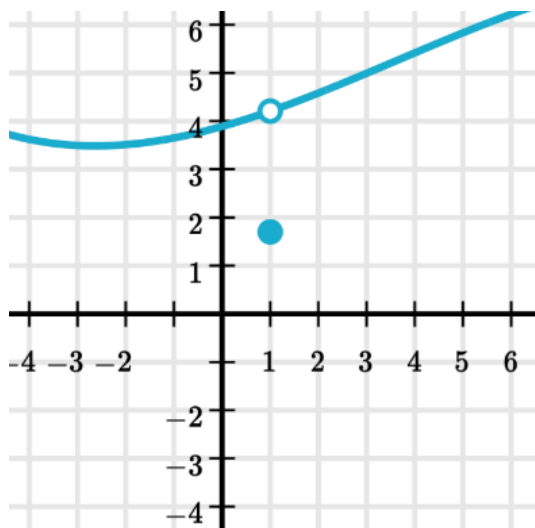
Find the equation of the tangent to the curve $y = 3x^2$ at $x = -1$. Write your answer in gradient-intercept form.

- (A) $y = -6x - 3$
- (B) $y = -6x + 3$
- (C) $y = 6x - 3$
- (D) $y = 6x + 3$
- (E) $y = -6x - 6$

Questions 7- 18 are short or long answer questions. Show all necessary **WORKING** and reasoning in the spaces provided. The value of each question is indicated. Part marks may be awarded.

Question 7 (1 marks) [Limits]

Consider the graph drawn below. Determine whether $\lim_{x \rightarrow 1} f(x)$ exists. Justify your answer.



Question 8 (4 marks) [Limits]

Evaluate the following, if they exist:

(a) $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 4}$

(b) $\lim_{x \rightarrow \infty} \frac{6x^3 - x + 2}{3x^2 + 7}$

Question 9 (2 marks) *[Derivative of powers of x]*

Find $\frac{dy}{dx}$ of $y = 2x + \frac{\sqrt{x}}{x}$, leaving your answer in surd form.

Question 10 (3 marks) *[Derivative of powers of x using rules 1-5]*

Find the gradient function of $f(x) = (x^2 - 1)(2x + 3)^4$ and simplify your answer fully.

Question 11 (3 marks) *[2nd derivative of any functions using rules 1-5]*

Find the second derivative of the function $y = \sqrt{x}(2 - \sqrt{x})$, writing your answer in surd form.

Question 12 (3 marks) *[Derivative at a point of any functions using rules 1-5]*

Find $f'(1)$ for $f(x) = (e^{-x^2+2})^3$ by using the chain rule. Leave your answer in exact form.

Question 13 (3 marks) *[Derivative of any functions using rules 1-5]*

Differentiate the function $y = \frac{\ln 2x}{x^2}$ and simplify your answer fully if possible.

Question 14 (3 marks) *[Derivative at a point after using rules for logarithms]*

“Note: You will not be given three questions in the exam. The three examples provided are to illustrate the different ways this question may be asked.”

Evaluate $f'(2)$ if $f(x) = \ln \frac{(2x-3)}{(x-1)}$.

OR Evaluate $f'(2)$ if $f(x) = \ln[(2x-3)(x-1)]$.

OR Evaluate $f'(2)$ if $f(x) = \ln\sqrt{(2x-3)}$.

Question 15 (3 marks) *[Derivative at a point of trigonometric function]*

For the function $f(x) = 2\pi \tan(2x)$, evaluate $f'\left(\frac{\pi}{6}\right)$. Give your answer as an exact value.

Question 16 (3 marks) *[Mixed derivative of any functions using rules 1-5]*

Find the derivative of $f(x) = \frac{e^{2x}}{\tan 2x}$ and simplify your answer if necessary.

Question 17 (3 marks) *[Equation of tangent to any function]*

Find the equation of the tangent to the curve $y = 2e^{\frac{x}{2}} + 1$ at the point where $x = 0$. Write the equation in gradient-intercept form.

Question 18 (3 marks) *[Application of tangent to function with powers of x]*

Note: You will not be given two questions in the exam. The two examples provided are to illustrate the different types of questions you may be asked.

Find the coordinates of the point on the curve $y = x^2 + 9x + 4$ where the tangent is parallel to the line $3x + y - 1 = 0$.

Or

Find the equation of the tangent to the curve $y = x^2 + 9x + 4$ at the point where the curve crosses the y -axis. Write the equation in general form.

END OF EXAM
List of Formulae

Rules for Differentiation:

Rule 1: *Constant multiplier rule:* $\frac{d}{dx}(kf) = kf'(x)$

Rule 2: *Sum rule:* $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$

Rule 3: *Product Rule:* $\frac{d(uv)}{dx} = u'v + uv'$

Rule 4: *Quotient Rule:* $\frac{d\left(\frac{u}{v}\right)}{dx} = \frac{u'v - uv'}{v^2}$

Rule 5: *Chain Rule:* $\frac{d}{dx}[f(g(x))] = f'(g(x)) \times g'(x)$

Derivatives:

If $y = x^n$ then $\frac{dy}{dx} = nx^{n-1}$

If $y = e^x$ then $\frac{dy}{dx} = e^x$

If $y = e^{f(x)}$ then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \ln x$ then $\frac{dy}{dx} = \frac{1}{x}$

If $y = \ln f(x)$ then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$ then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$ then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$ then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Equation of a Line:

Point-Gradient Form: $y - y_1 = m(x - x_1)$

Rules for Logarithms

$$\log_b xy = \log_b x + \log_b y$$

$$\log_b \left(\frac{x}{y} \right) = \log_b x - \log_b y$$

$$\log_b x^p = p \log_b x$$